

VIOLET GROVE, AB, Canada

WMO: 710620

Lat: 53.144N

Lon: 115.126W

Elev: 903

StdP: 90.94

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -28.7	(c) -25.5	(d) -32.5	(e) 0.2	(f) -28.4	(g) -29.2	(h) 0.3	(i) -25.2	(j) 10.4	(k) -0.9	(l) 9.0	(m) -2.1	(n) 2.1	(o) 250	(p) 0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.1	(c) 27.2	(d) 16.9	(e) 25.5	(f) 16.0	(g) 23.9	(h) 15.2	(i) 18.3	(j) 25.2	(k) 17.1	(l) 23.8	(m) 16.0	(n) 22.4	(o) 3.0	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.4	21.9	14.4	11.4	20.0	13.3	10.7	18.8	55.6	25.2	51.6	23.9	48.2	22.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.5	(c) 6.4	(e) -33.0	(f) 30.8	(g) 3.2	(h) 1.8	(i) -35.3	(j) 32.0	(k) -37.2	(l) 33.1	(m) -39.0	(n) 34.1	(o) -41.3	(p) 35.4		
(a) 9.0	(b) 7.5	(c) 6.4	(e) -33.1	(f) 20.0	(g) 3.2	(h) 1.1	(i) -35.4	(j) 20.8	(k) -37.3	(l) 21.5	(m) -39.1	(n) 22.1	(o) -41.4	(p) 22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-7.6	-7.1	-2.8	4.0	9.8	13.6	16.4	15.2	10.6	4.4	-2.2	-7.8
	DBStd	10.61	8.77	8.08	7.72	5.07	4.41	3.07	3.01	3.26	4.42	4.99	6.77	7.61
	HDD10.0	2816	546	478	396	186	58	4	1	3	44	182	368	552
	HDD18.3	5279	804	711	654	429	265	144	76	105	233	431	617	810
	CDD10.0	606	0	0	1	7	52	113	198	164	62	9	1	0
	CDD18.3	27	0	0	0	0	1	3	14	8	1	0	0	0
	CDH23.3	463	0	0	0	2	26	54	209	123	46	3	0	0
	CDH26.7	75	0	0	0	0	2	6	43	18	6	0	0	0
Wind	WSAvg	3.0	2.9	2.9	3.1	3.4	3.5	3.1	2.8	2.6	2.7	3.0	2.9	2.9
Precipitation	PrecAvg	549	26	19	19	32	63	95	106	70	42	28	24	15
	PrecMax	612	78	39	44	70	102	151	249	129	95	71	61	35
	PrecMin	448	2	3	3	7	28	28	45	28	10	7	6	4
	PrecStd	49	20	9	10	16	19	33	46	23	24	18	18	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	11.3	15.1	22.1	26.2	27.4	30.3	29.0	27.3	22.2	14.9	9.2
		MCWB	4.2	3.8	6.0	10.3	13.2	16.7	18.6	17.8	15.7	11.3	7.0	3.3
	2%	DB	7.8	7.8	11.2	18.2	23.3	24.8	27.7	26.4	24.5	17.3	11.1	6.6
		MCWB	2.7	2.0	3.9	8.4	12.2	15.0	17.9	16.9	14.7	9.3	5.0	1.8
	5%	DB	5.6	5.5	8.7	15.2	21.1	22.8	25.7	24.5	21.7	14.6	8.2	4.2
		MCWB	1.4	0.8	2.5	6.6	10.9	14.0	16.9	16.2	13.4	7.8	3.2	0.4
	10%	DB	3.4	3.3	6.6	12.5	18.7	20.9	23.7	22.5	18.5	12.0	5.6	2.1
		MCWB	0.1	-0.5	1.5	5.2	9.9	13.1	16.0	15.5	11.7	6.3	1.7	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	4.1	6.3	10.6	14.3	17.6	20.4	19.3	16.9	12.2	7.3	3.6
		MCDB	10.1	10.4	14.1	21.3	23.7	25.2	26.9	25.8	26.2	21.3	14.1	8.6
	2%	WB	2.9	2.4	4.2	8.9	12.7	16.0	18.8	18.0	15.1	9.7	5.3	2.0
		MCDB	7.1	7.3	10.5	17.5	21.6	22.9	25.8	24.8	22.9	16.6	10.4	6.1
	5%	WB	1.6	1.0	2.9	7.1	11.6	14.9	17.6	16.8	13.7	8.2	3.7	0.5
		MCDB	5.4	5.4	8.0	14.4	19.5	21.3	24.2	23.1	20.8	13.7	7.6	3.9
	10%	WB	0.2	-0.4	1.7	5.6	10.5	13.7	16.5	15.7	12.4	6.7	2.0	-0.9
		MCDB	3.2	3.0	6.2	11.6	17.6	19.5	22.7	21.6	17.8	11.2	5.3	1.9
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.2	11.2	11.4	12.3	11.5	12.1	12.0	11.8	10.4	8.8	9.0
		MCDBR	10.7	12.8	13.8	16.1	16.6	14.7	15.5	15.3	17.3	14.9	11.6	10.2
		MCWBR	7.0	8.2	8.2	8.6	8.0	7.1	7.2	7.5	8.6	8.3	6.9	7.0
	5% WB	MCDBR	10.6	12.5	13.0	15.4	14.9	13.0	14.3	14.3	16.1	14.3	11.1	10.0
		MCWBR	7.4	8.1	8.0	8.3	7.6	6.7	7.2	7.5	8.3	8.1	7.0	7.1
Clear-Sky Solar Irradiance	taub	0.246	0.256	0.285	0.326	0.338	0.350	0.348	0.355	0.312	0.280	0.253	0.236	
	taud	2.427	2.477	2.471	2.405	2.417	2.397	2.454	2.394	2.515	2.592	2.518	2.396	
	Ebn at Noon	777	883	916	909	911	900	896	867	874	843	768	717	
	Edh at Noon	52	68	85	103	108	111	104	104	81	61	47	44	
All-Sky Solar Radiation	RadAvg	0.97	1.96	3.40	4.62	5.51	5.86	6.07	5.03	3.50	2.04	1.04	0.72	
	RadStd	0.07	0.14	0.21	0.40	0.39	0.41	0.37	0.34	0.37	0.21	0.09	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-19
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page